STATEMENT OF PURPOSE

Applicant: Xuan Thanh Trinh
Program: Ph.D. in Electrical, Electronics and Computer Engineering
Institution: University of Michigan-Dearborn

Introduction

As a final-year student at Hanoi University of Science and Technology (HUST), my research focuses on distributed mathematical optimization for resilient energy management. While classical optimization provides strict physical guarantees, I recognize its computational limitations when scaling to real-time, dynamic grid operations. To overcome these bottlenecks, my overarching vision is to architect highly autonomous, scalable power systems. UM-Dearborn's Ph.D. program provides the exact collaborative framework necessary to develop next-generation control strategies for decentralized grids.

Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering instilled a systems-level perspective, training me to deconstruct complex power networks into dynamic, tractable optimization problems. This mathematical foundation drives my independent research trajectory, allowing me to architect solutions bridging abstract algorithms with real-world physical constraints.

In research submitted to *Sustainable Energy, Grids and Networks* (SEGAN), I developed a peer-to-peer trading framework subdividing distribution networks into autonomous microgrids, integrating grid physics (AC-OPF) with distributed optimization (ADMM). Validated across radial and ring-connected topologies, it minimized operational costs while strictly bounding voltages within the $\pm 5\%$ threshold and preserving public grid stability. However, this static framework lacked temporal foresight for sudden failures.

To achieve dynamic fault tolerance, my graduation thesis layered Model Predictive Control (MPC) over this baseline. By coordinating emergency power exchanges during simulated outages, this temporal control empowered neighboring networks to assist a failing grid, reducing forced load shedding from 15 MWh to just 3 MWh while protecting 100% of critical loads.

Forged by cross-group collaboration and rigorous convergence challenges, my physics-based modeling capabilities prepare me to drive scalable, real-time grid operations within your doctoral ecosystem.

Research Interests

My primary research objective centers on the comprehensive optimization of autonomous power systems. I focus on advancing economic dispatch and Unit Commitment (UC) for daily operational efficiency, while engineering dynamic network reconfiguration and rapid fault response for high survivability against extreme disruptions. These deterministic models guarantee strict operational constraints across both routine economic scheduling and critical self-healing interventions.

My secondary research integrates Artificial Intelligence to complement traditional grid operations.

My motivation lies in uncovering the untapped capabilities that emerge when fusing data-driven algorithms with physical models. To investigate this synergy, I target two Multi-agent Reinforcement Learning (MARL) applications: acting as a computational accelerator for instantaneous decentralized dispatch, and mitigating SOCP exactness issues alongside static thermal constraints.

Academic and Research Alignment

UM-Dearborn provides the rigorous theoretical coursework and collaborative environment necessary to pursue my synergistic vision. Specifically, my trajectory aligns directly with Prof. Van-Hai Bui's laboratory, driven by a shared focus on autonomous multi-agent energy management. Accordingly, I aim to advance his learning-based frameworks, such as FairMarket-RL and XRL-LLM, by bridging the gap between advanced AI capabilities and rigorous physical constraints.

Furthermore, the expertise of Dr. Wencong Su and Dr. Junho Hong strongly complements this holistic vision. Dr. Su's research utilizing LLMs for Economic Dispatch synergizes perfectly with my focus on system-level AI integration. Concurrently, Dr. Hong's expertise in physics-informed anomaly detection directly reinforces my approach to component-level physical constraints and grid security. Together, these cross-disciplinary insights will enable me to engineer highly resilient, autonomous grid architectures.

Career Objectives

Following my Ph.D., I intend to pursue a career as an R&D scientist dedicated to advancing global power infrastructures. I will apply these theoretical breakthroughs to solve high-impact challenges, driving the technical modernization and resilience of power grids worldwide.